1. List the key differences between IPv4 and IPv6 addressing by creating a two-column table with at least 4 points for each.

ANS:-

IPv4	IPv6
Uses a 32-bit address.	Uses a 128-bit address.
Written in dotted-decimal format, e.g., 192.168.1.1.	Written in hexadecimal, separated by colons, e.g., 2001:db8::1.
Provides approximately 4.3 billion addresses.	Provides an extremely large address space (\(2^{128}\) addresses).
Uses subnet masks such as 255.255.255.0.	Uses prefix notation such as /64.
Supports broadcast communication.	Does not use broadcast; it supports multicast and anycast.
NAT is commonly used to conserve public IPv4 addresses.	Provides a large address space, reducing the need for NAT for address conservation.

2. Given the following scenario: You have 4 devices (laptop, phone, smart TV, and printer) connected to your home WiFi. Assign a unique static IPv4 address to each device using the 192.168.10.0/24 network, and write down the IP, subnet mask, and default gateway for each device.

ANS:-

Network: 192.168.10.0/24

Subnet mask: 255.255.255.0

Default gateway: 192.168.10.1

Device	Static IP address	Subnet mask	Default gateway
Laptop	192.168.10.10	255.255.255.0	192.168.10.1
Phone	192.168.10.11	255.255.255.0	192.168.10.1
Smart TV	192.168.10.12	255.255.255.0	192.168.10.1
Printer	192.168.10.13	255.255.255.0	192.168.10.1

Explanation: Each device has a unique IP address within the same subnet. The default gateway allows the devices to communicate with networks outside their local subnet**.**

3. Calculate the number of valid subnets and hosts per subnet if you use a /27 CIDR subnet mask on a 192.168.1.0 network. List the first three valid subnet address ranges (network address to broadcast address) and the usable host IPs in each.

ANS:-

Given:

- Network: 192.168.1.0/24
- Required subnet mask: /27 = 255.255.255.224
- Borrowed host bits: \(27-24=3\)

Calculations:

- Number of subnets: \(2^3 = 8\)
- Host bits remaining: \(32-27=5\)
- Addresses per subnet: \(2^5=32\)
- Usable hosts per subnet: \(32-2=30\)
- Block size: \(256-224=32\)

The first three subnets are:

Subnet	Network address	Usable host IP range	Broadcast address
1	192.168.1.0	192.168.1.1 – 192.168.1.30	192.168.1.31
2	192.168.1.32	192.168.1.33 – 192.168.1.62	192.168.1.63
3	192.168.1.64	192.168.1.65 – 192.168.1.94	192.168.1.95

Answer: There are 8 subnets, each containing 30 usable host addresses, assuming the original network is 192.168.1.0/24.

4. You are given the IP address 10.0.5.17 with subnet mask 255.255.255.0. Determine the network address, broadcast address, and the range of valid host addresses for this subnet. Hint: Convert the subnet mask to CIDR notation and use bitwise AND for network address calculation.

ANS:-

Given:

- IP address: 10.0.5.17
- Subnet mask: 255.255.255.0
- CIDR notation: /24

Calculations:

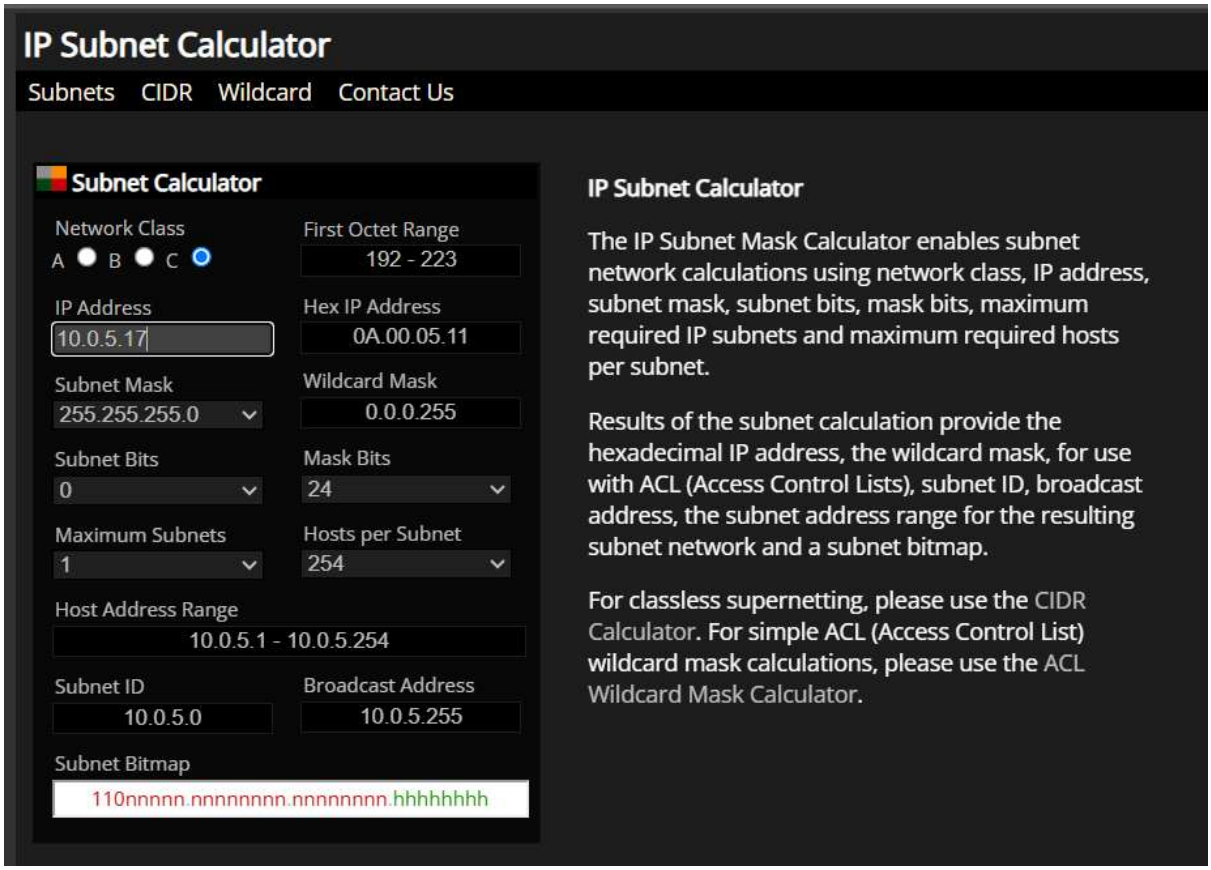
- Network address: 10.0.5.0
- Broadcast address: 10.0.5.255
- First usable host: 10.0.5.1
- Last usable host: 10.0.5.254
- Total usable host addresses: \(2^8-2=254\)

Explanation: The first 24 bits identify the network, while the remaining 8 bits identify hosts. Setting all host bits to 0 gives the network address, and setting them all to 1 gives the broadcast address.

Result: The valid host range is 10.0.5.1 to 10.0.5.254.

5. Use an online subnet calculator tool (such as https://www.subnet-calculator.com/) to verify your answer for the previous task. Take a screenshot of your calculation and write one line explaining if your manual answer matched the tool's output.

ANS:-



Observation: The expected output is:

- Network address: 10.0.5.0

- Broadcast address: 10.0.5.255
- Usable host range: 10.0.5.1 – 10.0.5.254

Conclusion: The manual subnetting calculation matches the expected output of the subnet calculator if it displays these same values.